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$N = 21, a = 2, b = 1, k = 4, \theta^\circ = 65^\circ, \alpha = 0{,}002$

No
Date

7) a. $S_n = 50 + (k \cdot n) + (-1)^n \cdot (a+1)$
$= 50 + (4 \cdot n) + (-1)^n \cdot (2+1) = 50 + 4n + (-1)^n \cdot 3$

n	S_n
1	$50 + (4 \cdot 1) + (-1)^1 \cdot 3 = 50 + 4 + (-3) = 51$
2	$50 + (4 \cdot 2) + (-1)^2 \cdot 3 = 50 + 8 + 3 = 61$
3	$50 + (4 \cdot 3) + (-1)^3 \cdot 3 = 50 + 12 + (-3) = 59$
4	$50 + (4 \cdot 4) + (-1)^4 \cdot 3 = 50 + 16 + 3 = 69$
5	$50 + (4 \cdot 5) + (-1)^5 \cdot 3 = 50 + 20 + (-3) = 67$
6	$50 + (4 \cdot 6) + (-1)^6 \cdot 3 = 50 + 24 + 3 = 77$
7	$50 + (4 \cdot 7) + (-1)^7 \cdot 3 = 50 + 28 + (-3) = 75$
8	$50 + (4 \cdot 8) + (-1)^8 \cdot 3 = 50 + 32 + 3 = 85$
9	$50 + (4 \cdot 9) + (-1)^9 \cdot 3 = 50 + 36 + (-3) = 83$
10	$50 + (4 \cdot 10) + (-1)^{10} \cdot 3 = 50 + 40 + 3 = 93$
11	$50 + (4 \cdot 11) + (-1)^{11} \cdot 3 = 50 + 44 + (-3) = 91$
12	$50 + (4 \cdot 12) + (-1)^{12} \cdot 3 = 50 + 48 + 3 = 101$

$S_n = \underbrace{(50 + 4n)}_{\text{tren linear}} + \underbrace{3(-1)^n}_{\text{osilasi}}$

Komponen tren: $T_n = 50 + 4n$, naik 4 setiap bulan
Komponen osilasi: $F_n = 3(-1)^n$, $+3$ jika n genap, -3 jika ganjil

karena selisih antar suku $(S_{n+1} - S_n)$ tidak konstan yg menyebabkan tidak murni aritmatika
$n = 1 = S_2 - S_1 = 61 - 51 = 10$
$n = 2 = S_3 - S_2 = 59 - 61 = -2$
dan juga ~~selisih~~ rasio S_{n+1} / S_n tidak konstan
$n = 1 = \frac{S_2}{S_1} = \frac{61}{51} = 1{,}196$

$n = 2 = \frac{S_3}{S_2} = \frac{59}{61} = 0{,}967$

No
Date

b) Hitung T_1 dan T_2

$T_1 = \sum_{n=1}^{12} S_n$

$S_n = 50 + 4n + 3(-1)^n$

$\sum_{n=1}^{12} 50 = 50 \cdot 12 = 600$

$\sum_{n=1}^{12} 4n = 4 . \frac{12 \cdot 13}{2} = 4 . 78 = 312$

$\sum_{n=1}^{12} 3(-1)^n = 3 \sum_{n=1}^{12} (-1)^n = 0$, karena 6 suku -1 dan 6 suku $+1$

$T_1 = 600 + 312 + 0 = 912$

$T_2 = \sum_{n=1}^{12} S_n^2$

$S_n^2 = [50 + 4n + 3(-1)^n]^2$ $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$

1. $a = 50, a^2 = 2500$ 'n sum $= 2500 \times 12 = 30.000$
2. $b = 4n, b^2 = 16n^2$ 'n sum $16 \sum n^2$, $\sum_{n=1}^{12} n^2$ $= \frac{12 \cdot 13 \cdot 25}{6} = \frac{3900}{6} = 650$
3. $c = 3(-1)^n, c^2 = 9$ 'n sum $= 9 \times 12 = 108$
4. $2ab = 2 \cdot 50 \cdot 4n = 400n$ 'n sum $400 \sum n = 400 \times 78 = 31200$
5. $2ac = 2 \cdot 50 \cdot 3(-1)^n = 300(-1)^n = 0$, karena 150 suku -1 dan 150 suku $+1$
6. $2bc = 2(4n) \cdot 3(-1)^n = 24n(-1)^n$

$24 \sum_{n=1}^{12} n(-1)^n = 24 . 6 = 144$

$n = 1$: -1 (-1)
$n = 2$: $+2$ (1)
$n = 3$: -3 $(-2$
$n = 4$: $+4$ (2)
$n = 5$: -5 (-3)
$n = 6$: $+6$ (3)
$n = 7$: -7 (-4)
$n = 8$: $+8$ (4)
$n = 9$: -9 (-5)
$n = 10$: $+10$ (5)
$n = 11$: -11 (-6)
$n = 12$: $+12$ (6)

No
Date

$G_n = G_1 \cdot r^{(n-1)}$

$G_1 = k = 4$

$r = 1 + (a/100) = 1 + 0{,}02 = 1{,}02$

$$S_{10}^{(\text{pelanggan})} = \sum_{n=1}^{10} G_1 r^{n-1} = G_1 \cdot \frac{r^{10} - 1}{r - 1}$$

$$= 4 \cdot \frac{(1{,}02)^{10} - 1}{0{,}02}$$

$1{,}02^2 = 1{,}0404$

$1{,}02^4 = (1{,}0404)^2 = 1{,}08243216$

$1{,}02^5 \approx 1{,}08243216 \times 1{,}02 = 1{,}1040808032$

$1{,}02^{10} \approx (1{,}1040808032)^2 \approx 1{,}218999442$

$(1{,}02)^{10} - 1 \approx 1{,}218999442 - 1 \approx 0{,}218999442$

$$S \approx 4 \times \frac{0{,}218999442}{0{,}02} = 4 \times 10{,}949721 = 43{,}798$$

$= 44$

Sekitar 44 pelanggan setelah 10 bulan

4. $\dfrac{(1{,}02)^n - 1}{0{,}02} > 500$

$\div 4$

$\dfrac{(1{,}02)^n - 1}{0{,}02} > 125$

$(1{,}02)^n - 1 > 2{,}5$

$(1{,}02)^n > 3{,}5$

$n \cdot \ln(1{,}02) > \ln 3{,}5$

$$n > \frac{\ln 3{,}5}{\ln 1{,}02} \approx \frac{1{,}25276}{0{,}0198026} \approx 63{,}25$$

n min = 64 bulan

F) Ya, residual cenderung:
- Positif untuk n genap
- Negatif untuk n ganjil

karena komponen osilasi $3(-1)^n$ pada data asli tidak ter-tangkap oleh regresi linear

Karena data memiliki musiman (seasonal) periodik setiap 2 bulan (+3, -3, +3, ...). Regresi linear hanya menangkap tren naik tetapi mengabaikan pola naik turun periodik.

Strategi perbaikan

1. Dekomposisi musiman : pisahkan tren + seasonal (dgn periode 2) + residual
2. Sarima $(0,1,1)(0,1,1)_2$ model musiman dgn periode 2
3. Regresi dgn variabel dummy untuk menunjukkan ganjil genap
4. Arima dengan X (regressor: $(-1)^n$).